Complete Trigonometry Crash Course

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Part I

The Basic Tools

Chapter 1

Geometry Tools Needed to Study Trigonometry

1.1 Right Triangles

There are three main types of angles:

- Acute Angles
- Right Angles
- Obtuse Angles

Definition 1 (Angles).
An **acute angle** is an angle with a measure less than 90°.
An **right angle** is an angle with a measure of 90°.
An **obtuse angle** is an angle with a measure greater than 90°.

Recall from geometry that you need three known measures to describe a triangle – the lengths of two sides and the measure of an angle, the lengths of three sides, or three other measures. For a right triangle, you need only two additional measures, because you already know that one of the angles' measure is 90°.

The following four cases describe all possible ways to describe a right triangle:

1. The lengths of two legs
2. The lengths of one leg and the hypotenuse
3. The lengths of one leg and the measure of an acute angle
4. The length of the hypotenuse and the measure of one acute angle

Suppose you need to calculate the lengths of all three sides. For cases 1 and 2, you only need geometry, specifically, the Pythagorean Theorem. But for cases 3 and 4, you need trig to get the remaining sides.

1.1.1 The Pythagorean Theorem

Let's take a look at the famous geometry tool *Pythagorean Theorem*, which helps us solve for cases 1 and 2.

Theorem 1 (Pythagorean Theorem)**.** The **Pythagorean Theorem** states that if a and b are the lengths of the legs of a right triangle and c is the length of the hypotenuse, then:

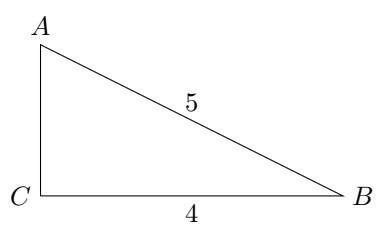
$$a^2 + b^2 = c^2$$

.

Let's illustrate how cases 1 and 2 can be solved with the help of the Pythagorean Theorem.

Sample Problem 1

Two legs of a right triangle measure 5 and 12 inches. Find the hypotenuse.



Since you know the length of the two legs, you can substitute those into the theorem's equation:

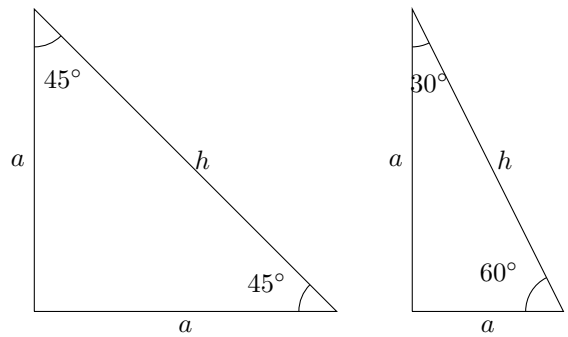
$$\begin{aligned} a^2 + b^2 &= c^2 \\ 5^2 + 12^2 &= c^2 \\ 25 + 144 &= c^2 \\ 169 &= c^2 \\ \sqrt{169} &= c \\ c &= 13 \end{aligned}$$

.

Solution: The length of the hypotenuse is 13 inches long.

1.1.2 Special Right Triangles

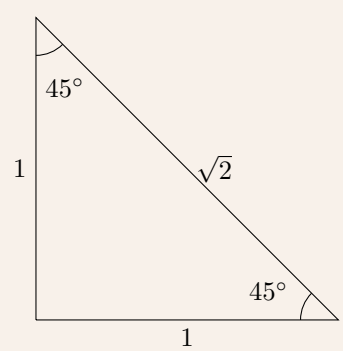
In this section, we'll take a look at two special kinds of right triangles. For these special triangles, if you know the length of any one side of the triangle, you can find the length of the other two sides.



The triangle on the left is called a $45° - 45° - 90°$ triangle, or an isosceles right triangle. The triangle on the right is called a $30° - 60° - 90°$ triangle. The Pythagorean Theorem is used to prove the following relationship that exist in these two special right triangles.

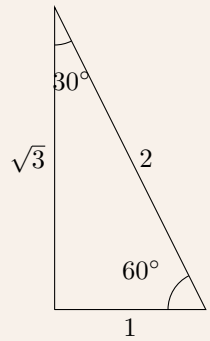
Definition 2 (Isosceles Right Triangle)**.** An **isosceles right triangle** is a triangle that has two equal sides and two equal angles, as well as a third angle that is a right angle. The only triangle that satisfies all these conditions is a $45° - 45° - 90°$ triangle.

Theorem 2. If each acute angle of a right triangle measures $45°$, the hypotenuse is $\sqrt{2}$ times as long as a leg.



Example. If one leg in a $45° - 45° - 90°$ triangle is 10 units long, then the hypotenuse is $10\sqrt{2}$ units long.

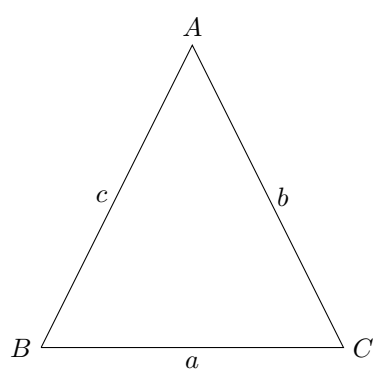
Theorem 3. If acute angles of a right triangle measure $30°$ and $60°$, then the hypotenuse is twice as long as the shortest leg and the longer leg is $\sqrt{3}$ times as long as the shortest leg.



Example. If the shortest leg in a $30° - 60° - 90°$ triangle is 6 units long, then the hypotenuse is $2 \times 6 = 12$ units long, and the longest leg is $6\sqrt{3}$ units long.

1.1.3 Classifying Triangles

Any triangle is six elements: three sides and three angles. Throughout this book, I use capital letters (A, B, C) to denote measures of angles and lowercase letters. I also use lowercase letters (a, b, c) to denote sides of the triangles I use the small letter corresponding to the name of the angle opposite to this side. I also use the symbol $\triangle$ to identity a triangle. Therefore, an expression $\triangle ABC$ means "the triangle ABC".



When given the lengths of three sides, you can construct a triangle that is determined by these sides; therefore, its angles are determined as well. However, you need to be careful thinking this way, because not all arbitrary side lengths

enable us to construct a triangle. Before constructing a triangle given its sides, you need to refer to the *Triangle Inequality*.

Definition 3 (Triangle Inequality)**.** The **triangle inequality** states that the sum of the lengths of any two sides of a triangle is greater than the length of the third side.

Example. Let's state whether it's possible for a triangle to exist with sides of the given lengths:

Sample 1: The given sides are 4, 5, and 6 units long. You need to compare the sums of any two sides with the third side. That means you need to exhaust all possibilities:

$$\begin{aligned} 4+5 &= 9 > 6 \\ 4+6 &= 10 > 5 \\ 5+6 &= 11 > 4 \\ &. \end{aligned}$$

Sample 2: The given sides are 1, 2, and 3 units long. Let's try all possibilities:

$$\begin{aligned} 2+3 &= 5 > 1 \\ 3+1 &= 4 > 2 \\ 1+2 &= 3 = 3 \\ &. \end{aligned}$$

The sum of these two sides (1 and 2) are not greater than the third side (3). Therefore, this triangle doesn't exist.

A triangle can be classified as acute (having three acute angles), right (having one right angle), and obtuse (having one obtuse angle). As we discussed, the lengths of the sides of a triangle determine its angles. Given the lengths of the sides, can you tell whether the triangle is acute, right, or obtuse?

The pythagorean theorem gives us a partial answer. We know that if the lengths satisfy the relationship $a^2 + b^2 = c^2$, then we know it's a right triangle. If it doesn't, then it's either an acute or obtuse triangle.

$$\begin{aligned} \text{If } \angle C \text{ of } \triangle ABC \text{ is acute, then } c^2 &< a^2 + b^2 \\ \text{If } \angle C \text{ of } \triangle ABC \text{ is obtuse, then } c^2 &> a^2 + b^2 \\ &. \end{aligned}$$

Using this information and given the sides' lengths, you can identify the type

of triangle you are dealing with.

Example. If the sides are 6, 7, 8, by substituting those values, you get:

$$8^2 = 64 < 6^2 + 7^2 = 36 + 49 = 85$$

.

So, this triangle is an acute triangle.

Note. In a triangle, the hypotenuse is always bigger in length than either of the sides. Hence, we always know which number should be substituted for c since it's the largest one of the three.

If the sides are 6, 8, and 10, then:

$$10^2 = 100 = 6^2 + 8^2 = 36 + 64 = 100$$

.

So the triangle is a right triangle.

If the sides are 5, 12, and 14, then:

$$14^2 = 196 > 5^2 + 12^2 = 25 + 144 = 169$$

.

So the triangle is an obtuse triangle.